

An uncertainty-aware safe-evolving reinforcement learning algorithm for decision-making and control in highway autonomous driving

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ABSTRACT

Rule-based and optimization-based approaches face challenges in decision-making and control for autonomous vehicles (AVs) in dynamic and complex highway scenarios. In contrast, reinforcement learning (RL) offers a more flexible and adaptable data-driven solution by allowing AVs to learn optimal actions through interactions with the environment, without requiring predefined rules or explicit programming. However, in real-world highway environments characterized by uncertainty, RL algorithm encounter difficulties in ensuring stability and safety. To address these challenges, this paper proposes an uncertainty-aware safe-evolving RL algorithm that integrates internal stability, external stability, and provable safety mechanisms. The internal stability mechanism ensures consistent performance improvements with high probability during policy updates itself, while the external stability leverages a benchmark policy as a reference to ensure the current policy performs at least as well as, if not better than, the benchmark. Furthermore, an action projection mechanism and a mixed learning procedure are incorporated to make minimal modifications to the learned policy, ensuring safety while supporting stable learning from both safe and original actions. The results show that the proposed algorithm maintains stability and safety throughout the learning process, achieves final performance comparable to traditional RL methods, and delivers higher training efficiency in a complex dynamic highway scenario in simulation. This suggests that the algorithm offers a viable solution for self-evolving systems in uncertain real-world environments, where traditional approaches may struggle.

1. Introduction

Autonomous vehicles (AVs) enhance safety, improve traffic efficiency, and increase accessibility, offering a more convenient and sustainable transportation solution (Hu et al., 2023). Among all tasks in AVs, tackling the decision-making and control task in highway scenarios is crucial, as highways involve high speeds, complex traffic flows, and varying driver behaviors, requiring accurate and timely decisions to ensure safe navigation, smooth lane merging, and efficient travel while minimizing the risk of accidents (Zhang et al., 2025).

Reinforcement learning (RL) has shown great potential in addressing these challenges, enabling AVs to learn optimal policies through exploration and feedback, adapt to dynamic environments, and make real-time decisions without relying on predefined rules or extensive programming (Hu et al., 2024). However, most RL applications cannot be directly trained in real-world scenarios and still depend heavily on

simulations, necessitating the development of a high-fidelity simulation model prior to policy learning. This is because in the *real-world* autonomous driving (AD) environment characterized by uncertainty, RL algorithm still faces two major challenges: instability during online learning and the difficulty of ensuring system safety throughout the learning process.

Instability during online learning is a core challenge for self-evolving RL systems, often closely linked to the epistemic uncertainty of policies in real-world environments. This uncertainty can introduce biases during policy evaluation, leading to fluctuations in decision-making and further exacerbating performance instability. In real-world settings, such instability can also lead to potential safety risks, making RL systems seem untrustworthy and ultimately hindering their deployment. Consider how a human learns to drive steadily in uncertain environments: They progressively refine their driving skills over time, cautiously ensuring consistent improvement while avoiding mistakes or

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inconsistencies in steering, braking, or acceleration, and they evaluate their driving against external benchmarks, such as instructor feedback or driving skill videos, using these references to guide their improvement. Mathematically, human driving behaviours align with internal and external learning stability. Internal stability focuses on the update stability of the decision-making and control policy itself, as depicted in Fig. 1. Its objective is to guarantee, with high probability, consistent performance improvements, ensuring the policy evolves and improves over time. In contrast, external stability involves evaluating the learned policy against an external benchmark, as depicted in Fig. 2. The goal is not for the policy to exactly replicate the benchmark, but to use it as a stable reference, ensuring that the policy consistently outperforms the benchmark. While integrated internal and external stability may support stable learning for real-world autonomous driving, no existing literature, to the authors' knowledge, combines these mechanisms.

In addition, although stable learning mechanisms can improve system stability during real-world policy learning, they alone cannot ensure the safety of AVs throughout the learning process. This is primarily because RL relies on a reward-driven mechanism, which lacks hard constraints to ensure safety. Given that any unsafe action could result in severe consequences for AVs, it underscores the importance of incorporating additional safety measures throughout the learning process (Yang et al., 2024). Traditional safe RL methods can help the system achieve relatively safe policies. However, these methods typically reduce unsafe behaviours under specific conditions and cannot provide absolute safety guarantees (Yang et al., 2025). In contrast, provably safe RL methods offer explicit safety assurances by formally verifying that the system's actions will not lead to unsafe outcomes, ensuring that the system operates within predefined safety constraints at all times. This approach provides a higher level of confidence in the safety of the system, particularly in complex or high-risk highway environments. As summarized by Hanna Krasowski et al., provably safe RL methods can be categorized into action masking, action replacement, and action projection (Krasowski et al., 2023). Action masking limits the agent to selecting actions only from a predefined safe action set. While it has been applied to simple control (Fulton and Platzer, 2018) (Brosowsky et al., 2021) and decision-making tasks in AD (Mirchevska et al., 2018) (Krasowski et al., 2020) (Krasowski et al., 2022), its application to complex decision-making and control tasks remains challenging, due to the significant difficulty of computing safe action sets in complex scenarios. Action replacement is another approach that replaces unsafe actions chosen by the agent with safe ones, providing a practical solution for the decision-making and control task in AVs (Evans et al., 2023) (Wang et al., 2023). However, significant differences between the actions proposed by RL and the replaced actions can disrupt control



Fig. 1. Internal stability schematic diagram.

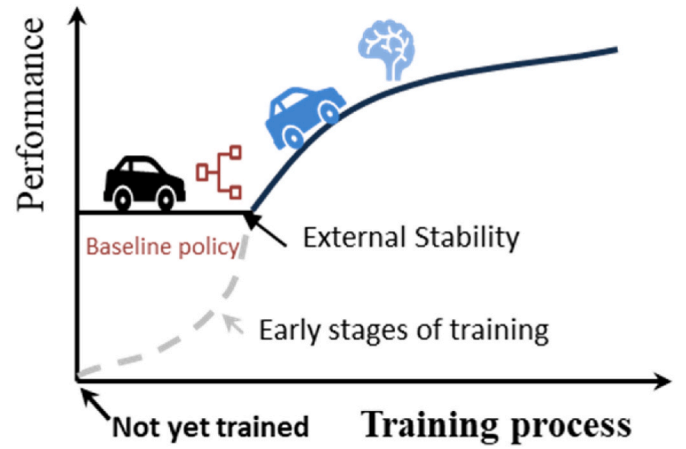


Fig. 2. External stability schematic diagram.

continuity, causing erratic or unpredictable vehicle behavior. Compared to action replacement, action projection aims to select the closest safe action to the one proposed by RL, avoiding the issues caused by large discrepancies in action replacement (Kochdumper et al., 2023). However, a key challenge arises from the discrepancy between the training data distribution and the actual policy distribution when directly trained in the real world. Specifically, the learned policy is trained on samples from the mixed behavior policy that includes both RL-generated actions and actions enforced by the safety shield. This violates the assumptions of standard off-policy RL algorithms which rely on the data matching the learned policy distribution, leading to increased errors in critic estimation and instability in policy learning.

To tackle the stability and safety challenges of online learning in self-evolving RL systems, this paper proposes a novel framework integrating internal stability, external stability, and a provably safe action layer. Internal stability ensures consistent performance improvements during policy updates itself. Building on this, external stability leverages a benchmark policy as a reference to ensure the current policy performs at least as well as, if not better than, the benchmark. Furthermore, an action projection method provides hard safety guarantees by correcting any unsafe actions. To resolve the data distribution mismatch caused by such interventions and enable the agent to learn from them, we introduce a crucial mixed learning procedure. The synergistic combination of these components enables continuous, safe, and robust online learning for AD systems. Specifically, the contribution of this paper are as follows:

- (1) To combat the inherent instability of policy updates in traditional RL, we propose an internal stability mechanism. This mechanism ensures that each policy update itself achieves a consistent performance improvement with high probability, thereby preventing the drastic performance fluctuations common during training. Furthermore, by integrating a critic ensemble with a gradient diversification objective, our approach also significantly enhances computational efficiency.
- (2) To mitigate the high uncertainty that plagues the early stages of training, we introduce an external stability mechanism. This mechanism anchors the learning process by referencing a benchmark policy, guaranteeing that the agent's performance will not degrade below this established benchmark. This provides reliable guidance during initial exploration and demonstrably accelerates the convergence towards an optimal and robust policy.
- (3) To address the safety concerns during the learning process, we employ a provably safe RL mechanism to ensure safety throughout training. This is combined with a mixed learning procedure to resolve the potential stability issues arising from the

discrepancy between the mixed training data distribution and the actual policy distribution for real-world AD learning. This approach steers the policy towards continuous and safe improvements, ultimately achieving a policy that is both safe and high-performing.

The rest of this article is organized as follows. Section 2 provides a review of related works in safe reinforcement learning. Section 3 presents the problem formulation for safe self-evolving decision-making and control for AVs. Section 4 presents the details of the proposed uncertainty-aware safe-evolving RL algorithm. Section 5 details the experimental setup and discusses the results, including a comparison with baseline methods. Finally, Section 6 concludes this article. The abbreviations used in the study are listed in Table I.

2. Related works

Safe reinforcement learning methods can be broadly divided into two categories: model-based and model-free approaches. Model-based methods leverage an explicit dynamics model of the environment to plan and verify safe actions, whereas model-free methods learn safe policies directly from data through interaction. Since our RL algorithm is based on a model-free approach, this section will focus exclusively on reviewing prior work within the model-free safe RL paradigm, with a particular emphasis on its applications in autonomous driving.

2.1. Safe RL methods

Within the model-free paradigm, various techniques have been developed to enforce safety. These can be broadly categorized based on their underlying mechanism for incorporating safety constraints, primarily including policy optimization, methods using Gaussian Processes (GPs), and approaches based on formal methods. Policy optimization-based approaches: These methods typically formulate the safe RL problem as a constrained optimization problem, often within a policy gradient framework. A seminal work in this area is Constrained Policy Optimization (CPO) (Achiam et al., 2017), which guarantees that safety constraints are satisfied at every policy update by using a trust region. CPO ensures monotonic reward improvement while satisfying constraints, but its reliance on second-order optimization and the fisher information matrix can be computationally expensive. Building on this, Projection-Based Constrained Policy Optimization (PCPO) (Yang et al., 2020) proposes a two-step approach: first, it performs an unconstrained update to maximize the reward, and then projects the updated policy back into the set of safe policies. This simplifies the process but may still suffer from approximation errors and sampling inefficiency, especially

when the policy moves far from the safe set. Gaussian processes-based approaches: GPs are a powerful non-parametric tool for safe exploration by explicitly modeling uncertainty. In safe RL, GPs are often used to model unknown reward or cost functions. For instance, Sui et al. (2015) use a GP to model the unknown objective function and leverage its confidence bounds to guide exploration, ensuring that the agent avoids unsafe regions with high probability. Similarly, Turchetta et al. (2016) use GPs to approximate unknown transition dynamics in finite MDPs, enabling safe exploration. The primary advantage of GP-based methods is their ability to quantify uncertainty, which is crucial for balancing exploration and safety. However, their main limitation is that they typically provide probabilistic safety guarantees rather than absolute, deterministic ones. Formal methods-based approaches: Formal methods leverage tools from logic and automata theory to provide rigorous, provable safety guarantees. In the context of safe RL, this often involves specifying safety rules using formal languages like Linear Temporal Logic (LTL). Hasanbeig et al. (2020) propose an LTL-based approach where logical formulas act as hard constraints during policy synthesis, ensuring the agent's behavior adheres to complex specifications. Another approach by Murugesan et al. (2019) uses satisfiability modulo theories solvers as a verification layer to check and enforce safety. The strength of these methods lies in their ability to provide hard safety guarantees. However, the primary limitation of formal methods is the requirement for an explicit, mathematically precise model of the system and its environment, coupled with the often-high computational complexity of verification.

2.2. Safe RL methods for autonomous driving

While the general machine learning methods discussed above have proven valuable in many related automotive domains, of such as driver monitoring and energy management (Khawaja et al., 2023) (Alzubi et al., 2022) (Feng et al., 2025), their application to autonomous driving operates under far more stringent safety requirements. In a high-stakes domain like autonomous driving, methods limited to soft constraints (as in some policy optimization techniques) or only probabilistic guarantees (common in GP-based approaches) are often inadequate for real-world deployment. This necessity has driven the adoption of methods that offer strict, verifiable safety guarantees. These methods, collectively known as Provably Safe RL, are conceptually rooted in the aforementioned formal methods. In practice, they are typically realized by integrating a supervisory layer that performs safety verification and intervenes at the action level. These practical implementations can be broadly categorized into three paradigms: action masking, action replacement, and action projection (Krasowski et al., 2023). Table II provides a summary and comparison of representative works within each of compared. Action masking pre-emptively restricts the agent's policy to a set of verified safe actions. In the context of autonomous driving, this has been applied to high-level lane-changing decisions (Mirchevska et al., 2018) (Krasowski et al., 2020) and to ensure safety in Adaptive Cruise Control (ACC) by imposing state-specific safe sets (Brosowsky et al., 2021). Its main drawback, however, is the computational difficulty of identifying the complete safe action set in real-time for continuous control tasks. Action replacement ensures safety via a supervisory module that substitutes any unsafe action from the RL agent with a pre-defined safe one. This paradigm has been applied in various driving scenarios. For highway lane-changing, for instance, one approach integrates a regret-based human driver model to inform safe actions (Chen et al., 2020), while another proposes the Simplex-Drive framework, which dynamically switches to a safe baseline controller to prevent collisions (Chen et al., 2022). For high-speed autonomous racing, Evans et al. (2023) employed a viability theory-based supervisor to guarantee safety. While robust, the primary drawback of this approach is the discrepancy between the agent's intended action and the replacement, which can disrupt the learning process and cause control discontinuity. In contrast, action projection offers a more refined

Table I
Abbreviations used in the study.

Abbreviation	Full form
ACC	Adaptive Cruise Control
AD	Autonomous Driving
AVs	Autonomous Vehicles
CBFs	Control Barrier Functions
CPO	Constrained Policy Optimization
EDAC	Ensemble-Diversified Actor Critic
GPs	Gaussian Processes
LCB	Lower Confidence Bound
LTL	Linear Temporal Logic
MDP	Markov Decision Process
ORCA	Optimal Reciprocal Collision Avoidance
PCPO	Projection-Based Constrained Policy Optimization
PLS	Polynomial Level Set
RL	Reinforcement learning
RSS	Responsibility-Sensitive Safety
SAC	Soft Actor-Critic
SAC-N	Soft Actor-Critic with N
SPOT	Set-Based Prediction Of Traffic Participants

Table 2
Comparison of provably safe RL methods in autonomous driving.

References	Verification Method	Action space	Application scenarios
Action masking			
Mirchevska et al. (Mirchevska et al., 2018)	Maneuver trajectory verification against rule-based safe sets	discrete	highway
Krasowski et al. (Krasowski et al., 2020)	Reachability-based collision verification using set-based predictions	discrete	highway
Brosowsky et al. (Brosowsky et al., 2021)	Derivation of safe action intervals from a Responsibility-Sensitive Safety (RSS) model.	continuous	ACC
Action replacement			
Chen et al. (Chen et al., 2020)	Collision prediction verification based on a regret-theory human behavior model	discrete	Highway
Chen et al. (Chen et al., 2022)	Verified safe controller using Optimal Reciprocal Collision Avoidance (ORCA)	continuous	Highway
Evans et al. (Evans et al., 2023)	Offline computation of a recursively feasible safe set	continuous	racing
Action projection			
Kochdumper et al. (Kochdumper et al., 2023)	set-based reachability analysis	continuous	highway
Cheng et al. (Cheng et al., 2019)	CBFs	continuous	Car-following
Wang et al. (Wang, 2022)	CBFs	continuous	highway

solution by correcting an unsafe action to the closest point within the safe set. This paradigm has been successfully applied to various autonomous driving scenarios using different formal methods. For instance, set-based reachability analysis has been employed to ensure safety in general highway driving (Kochdumper et al., 2023). Another prominent technique, Control Barrier Functions (CBFs), has also been used effectively, with applications demonstrated in both autonomous car-following (Cheng et al., 2019) and more complex highway driving environments (Wang, 2022). The key advantage of action projection is that it makes minimal modifications to the agent's learned policy. This preserves the agent's original intent as much as possible, thereby maintaining better control continuity and learning stability compared to the other methods. However, a critical challenge remains in how to guide the agent to learn from these safety interventions.

3. Problem formulation

This study examines an uncertainty-aware safe-evolving RL algorithm in the context of a three-lane highway, where frequent lane changes, varying vehicle speeds, and unpredictable driving conditions, as depicted in Fig. 3. The algorithm should demonstrate strong initial performance in real-world scenarios by leveraging its understanding of the epistemic uncertainty observed in a relatively well-behaved

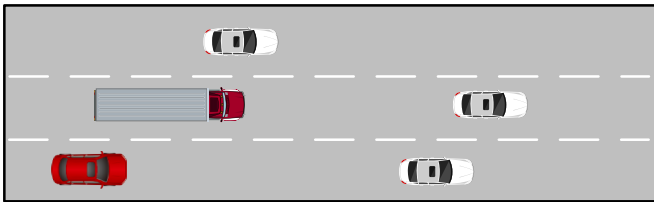


Fig. 3. Schematic diagram of driving scenario with the ego vehicle (red) and surrounding traffic participants. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

simulation environment. Building on this foundation, the goal of this research is to enable a safe self-evolving RL process in real-world environments.

Specifically, the decision-making and control problem can be formulated as an infinite-horizon Markov Decision Process (MDP), defined as a 5-tuple $M = \langle S, A, P, R, \gamma \rangle$, where S is the state space, A is the action space, R is the reward function, P is the state transition kernel, γ is the discount factor, which balances short-term and long-term rewards. The policy π is a function that maps states from the state space to actions, determining the action to be taken for each state. The state-value function $V_\pi(s)$ represents the expected cumulative reward when starting from state s and following policy π , expressed as:

$$V_\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^T \gamma^t r_t \mid s_0 = s \right] \quad (1)$$

The action-value function $Q_\pi(s, a)$ represents the expected cumulative reward of taking action a in state s and then following policy, defined as:

$$Q_\pi(s, a) = \mathbb{E}_\pi \left[\sum_{t=0}^T \gamma^t r_t \mid s_0 = s, a_0 = a \right] \quad (2)$$

where T denotes the total number of time steps considered. The expected performance of a policy π is defined as $(\pi) = \mathbb{E}_{s \sim \rho} V_\pi(s)$, where r_t is the immediate reward at time step t , and ρ is the initial state distribution. In general, the objective of a RL-based problem is to find an optimal policy π through multiple rounds of training in simulation environments, such that the expected cumulative reward $J(\pi)$ is maximized under the distribution ρ . However, traditional RL methods, which rely on simulation-based training followed by real-world deployment, often lack robustness when faced with uncertain real-world conditions. A future promising solution involves adopting a self-evolution RL paradigm in real-world environments, where continuous adaptation and improvement occur based on real-time data and interactions.

While this approach removes the reliance on perfect simulation environments, implementing the self-evolution of RL algorithms in real-world settings presents significant challenges, particularly in ensuring stability. This is because the balance between exploration and exploitation in RL algorithms can lead to short-term performance degradation when attempting new policies. Moreover, instability in the policy update process, such as improper learning rate settings or limitations of optimization methods, can result in significant variations, further destabilizing performance. Finally, starting from scratch without prior knowledge makes it difficult for the algorithm to adapt stably to complex environments in the early stages, leading to performance fluctuations.

Furthermore, safety is another critical issue throughout the learning process in the real-world environments. This is due to the fact that RL relies on a reward-driven mechanism that lacks hard constraints to guarantee safety. While adopting safety layers (e.g., safety shields) could potentially modify RL-generated actions for safety after the training is complete, directly applying them during training introduces a new challenge: the discrepancy between the training data distribution and the actual policy distribution. This mismatch can destabilize the learning process, affecting both performance and safety.

4. Approach overview

In Section 4, we propose a safe evolving RL framework that integrates internal stability, external stability, and a safety shield to enable the stability and safety of AD policy updates in the real world. Targeting AD's decision-making and control strategy that outputs throttle and steering angle commands, this framework should meet the requirements for safety (avoiding collisions), efficiency (increasing driving speed), and comfort (reducing frequent lane changes) to adapt to the dynamic

and uncertain nature of highway lane-changing scenarios. Section 4.1 provides a detailed explanation of the internal stability mechanism, aimed at improving the stability of policy updates itself. Section 4.2 elaborates on the external stability mechanism, which leverages a benchmark policy as a reference to ensure the current policy performs at least as well as, or better than, the benchmark. Section 4.3 describes the implementation of a safety shield and a mixed learning procedure after the safety shield is enforced. Finally, Section 4.4 outlines the specific experimental setup.

4.1. Internal stability (self-comparison)

Ensuring the stability of policy updates itself is critical in optimizing decision-making and control policy for AD, particularly in real-world environments where uncertainty, and safety risks are heightened. Unstable policy updates can lead to uncontrollable situations, jeopardizing the operational safety and performance of AVs. Soft Actor-Critic (SAC) (Haarnoja et al., 2018), a widely used advanced RL algorithm, has demonstrated relatively stable performance in simulation environments while effectively handling a variety of tasks. To achieve this, SAC employs a core mechanism involving critic and target critic networks. The critic network evaluates the value of state-action pairs in real-time, while the target critic network, updated at a slower rate, ensures stability in the learning process and reduces policy fluctuations.

While SAC demonstrates strong performance in simplified simulation settings, it often struggles in dynamic, complex, and uncertain real-world scenarios. These challenges are largely attributed to epistemic uncertainty, which stems from the model's limited capacity to generalize to unseen or out-of-distribution state-action pairs. This uncertainty can lead to a cascade of problems, including biased value estimates and unstable policy updates, which in turn result in ultimately hinder the algorithm's effectiveness. To tackle this issue, G. An et al. (2021) proposed a feasible approach using the SAC-N method to quantify uncertainty. Specifically, SAC-N introduces an ensemble of critic networks, aggregating the evaluation results of multiple Q-networks to provide more conservative Q-value estimates. This approach effectively prevents the selection of suboptimal actions due to Q-value overestimation, offering significant support for robust policy design in real-world environments.

However, in the complex physical world, SAC-N still faces two significant challenges. Firstly, SAC-N's reliance on a large ensemble of Q-networks introduces high computational overhead, which limits its efficiency in practical applications. Secondly, SAC-N relies on traditional periodic updates for target network, a mechanism that poses limitations in dynamic environments. In detail, if the policy update period is too large, it can lead to stale updates, reducing the effectiveness of learning and potentially destabilizing the training process. Conversely, overly frequent updates can lead to excessive adjustments, compromising overall stability and performance. To overcome the aforementioned challenges, we present an internal stability mechanism that efficiently handles stability issues arising between successive policy updates, utilizing two key techniques. Firstly, drawn inspiration from Ensemble-Diversified Actor Critic (EDAC) (An et al., 2021), a gradient diversification objective $J_{ES}(Q_\phi)$ to enhance the diversity within the Q-network ensemble is incorporated in the original SAC-N framework. By minimizing $J_{ES}(Q_\phi)$, the alignment between the gradients of different Q-networks is reduced, encouraging a more diverse gradient distribution. The increased gradient diversity reduces the necessity for a large ensemble, enabling fewer Q-networks to achieve performance similar to that of a larger ensemble, which significantly lowers computational complexity.

$$J_{ES}(Q_\phi) := \mathbb{E}_{s,a \sim D} \left[\frac{1}{N-1} \sum_{1 \leq i \neq j \leq N} \langle \nabla_a Q_{\phi_i}(s, a), \nabla_a Q_{\phi_j}(s, a) \rangle \right] \quad (3)$$

where $\nabla_a Q_{\phi_i}(s, a)$ represents the gradient of the i -th Q-network with respect to the action a , and $\langle \cdot, \cdot \rangle$ denotes the cosine similarity between gradients.

In addition, to guide policy updates in the direction of progressively better performance and also take epistemic uncertainty into account, we refine the periodic update strategy according to whether the Lower Confidence Bound (LCB) of the current policy's Q-values is better than those of the target critic network with high probability, as formally expressed by:

$$\mathbb{P}(Q_\pi^-(s, a) \geq Q_{target}^-(s, a)) \geq 1 - \delta \quad (4)$$

To achieve internal stability, as illustrated in Fig. 4, the ensemble critic network and the ensemble target critic network evaluate policy performance by computing the LCB of the state-action values. During each update cycle, n state-action pairs are sampled from the replay buffer, and their respective LCBs are calculated and compared to determine if the current policy demonstrates a confident improvement over the target. For each sampled state-action pair (s_i, a_i) , the critic and target critic ensembles each provide a set of N Q-value estimates, denoted as $\{Q_\pi^{(j)}(s_i, a_i)\}_{j=1}^N$ and $\{Q_{target}^{(j)}(s_i, a_i)\}_{j=1}^N$ respectively. For each ensemble, we compute the mean (μ) and standard deviation (σ) of these estimates. For the critic network (and similarly for the target critic network), these are:

$$\mu_\pi(s_i, a_i) = \frac{1}{N} \sum_{j=1}^N Q_\pi^{(j)}(s_i, a_i) \quad (5)$$

$$\sigma_\pi(s_i, a_i) = \sqrt{\frac{1}{N} \sum_{j=1}^N (Q_\pi^{(j)}(s_i, a_i) - \mu_\pi(s_i, a_i))^2} \quad (6)$$

Using a specified confidence level δ (e.g., $z_\delta \approx 1.645$ for δ 95% confidence level), the LCBs for the current policy (Q_π^-) and the target policy (Q_{target}^-) are expressed as:

$$Q_\pi^-(s_i, a_i) = \mu_\pi(s_i, a_i) - z_\delta \frac{\sigma_\pi(s_i, a_i)}{\sqrt{N}} \quad (7)$$

$$Q_{target}^-(s_i, a_i) = \mu_{target}(s_i, a_i) - z_\delta \frac{\sigma_{target}(s_i, a_i)}{\sqrt{N}} \quad (8)$$

After obtaining the LCBs for all state-action pairs in the batch, a significance test is performed to determine if $Q_\pi^-(s_i, a_i)$ is superior to $Q_{target}^-(s_i, a_i)$. We define n_{sig} as the count of pairs that satisfy this condition:

$$n_{sig} = \sum_{i=1}^n \mathbf{1}\{Q_\pi^-(s_i, a_i) \geq Q_{target}^-(s_i, a_i)\} \quad (9)$$

Where $\mathbf{1}\{\cdot\}$ is the indicator function., which equals 1 if $Q_\pi^-(s_i, a_i) \geq Q_{target}^-(s_i, a_i)$ and 0 otherwise. Finally, we calculate the proportion of pairs, $p: p = \frac{n_{sig}}{n}$. If this proportion p is greater than or equal to a preset confidence threshold $1 - \delta$ (i.e., $p \geq 1 - \delta$), we conclude that the current policy demonstrates a confident improvement over the target across the batch, which triggers an update of the target network. By making the update of the learning target conditional on this rigorous statistical test, we prevent the agent from chasing a noisy or fluctuating objective. This ensures the learning "goalposts" only move when we are highly confident of genuine progress, thereby stabilizing the entire learning process.

4.2. External stability (comparison with benchmark policy)

While internal stability ensures stability during the learning process through self-comparison, starting from scratch without prior knowledge introduces high epistemic uncertainty in the early stages due to limited experience and data, potentially causing notable performance varia-

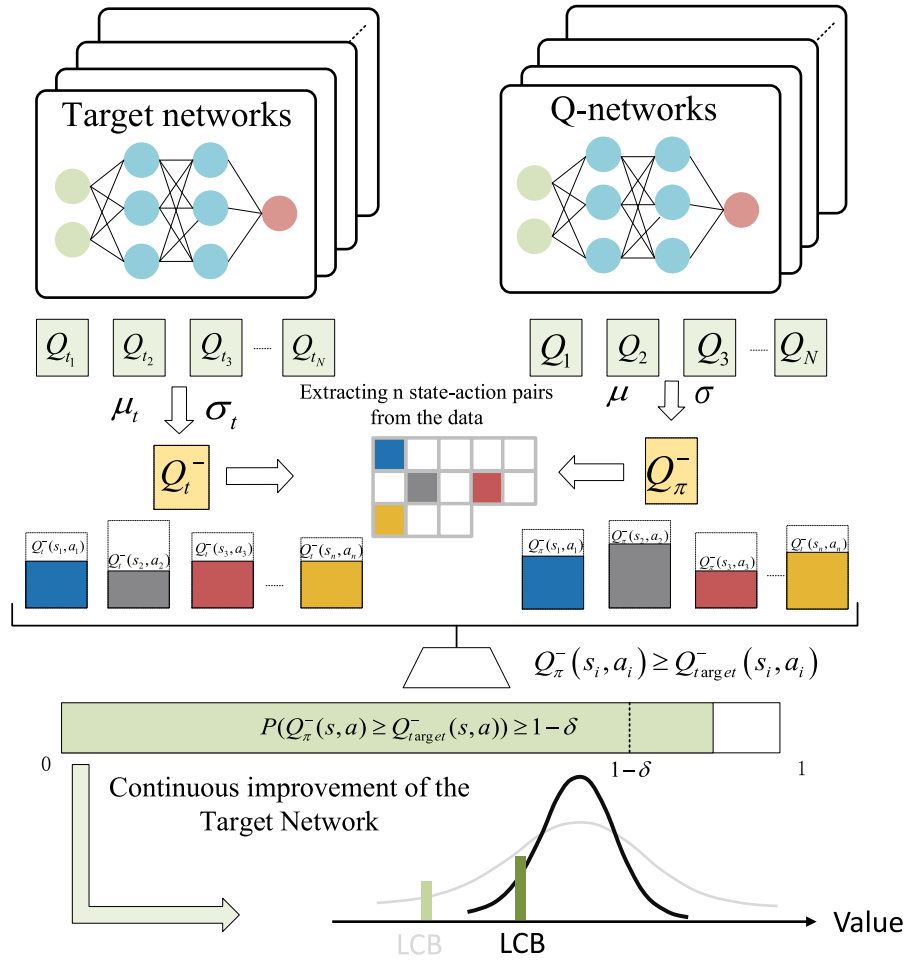


Fig. 4. Illustration of internal stability.

tions. To resolve this issue, we introduce external stability by restricting the agent to an admissible action set comprising actions that are highly probable to outperform a benchmark policy. Our external stability mechanism is grounded in the theoretical principle that a policy constructed by selecting the best action from a set of existing policies will have a value greater than or equal to the value of any individual policy in that set. We can formalize this as a theorem:

Theorem 1. Given a set of M policies $\{\pi^{(i)}\}_{i=1}^M$ and define $\bar{\pi}$ such that for all $s \in \mathcal{S}$, $\bar{\pi}(s) \in \arg \max_{i \in [M]} v^{\pi^{(i)}}(s)$. Then,

$$v^{\bar{\pi}}(s) \geq \max_{i \in [M]} v^{\pi^{(i)}}(s), \forall s \in \mathcal{S}. \quad (10)$$

Our external stability mechanism is a practical, step-wise implementation of this principle, where the set of policies effectively consists of two: the current learned policy π and the trusted benchmark policy π_b .

To be specific, the benchmark policy π_b is firstly trained in a relatively well-behaved simulation environment, which also employs the principles of ensemble networks, to provide a relatively stable and conservative initial performance. After that, the action space for high-probability stable improvement is identified based on whether the LCB of the current policy's Q-value is greater than or equal to the LCB of the benchmark policy's V-value, ensuring that the current policy performs at least as well as, or better than, the benchmark policy, as illustrated in Fig. 5. The action space for high-probability stable improvement is formally expressed as:

$$\mathcal{A}(s) = \{a \in \mathcal{A} : Q_\pi^-(s, a) \geq V_{\pi_b}^-(s)\} \quad (11)$$

where Q_π^- represents the LCB of Q_π , and $V_{\pi_b}^-$ represents the LCB of V_{π_b} , both evaluated using the ensemble network. For the benchmark policy's LCB $V_{\pi_b}^-$, given that the action space is continuous, a fixed number of actions are sampled from π , and each action is evaluated to construct the high-probability stable improvement action space $\mathcal{A}(s)$.

During the early stages of learning, as the LCB of the current policy is relatively low, no action can satisfy (11) and thus the system primarily relies on the benchmark policy π_b . As data accumulates and the current policy improves, the LCB of the current policy gradually increases, allowing the system to transition decision-making from the benchmark policy to the current policy, thereby enhancing adaptability from simulation to real-world environments.

4.3. Integrating safety shields in RL training

Relying solely on the previously mentioned internal and external stability mechanisms is not enough to fully guarantee the safety of AVs during real-world learning, because RL is driven by rewards and safety cannot be entirely ensured through rewards alone. To enable safe self-evolving learning in real-world environments, a safety shield with an action projection mechanism is implemented to maintain safety during training. In addition, a mixed learning procedure is introduced to address the potential stability issues caused by the mismatch between the mixed training data distribution and the actual policy distribution.

The core idea of the safety shield, as illustrated in Fig. 6, is to compute the reachable set of the ego vehicle over a time horizon t_f , representing all states the vehicle can achieve under given control input constraints $u(t) \in U$. Unlike optimizing a single control input trajectory,

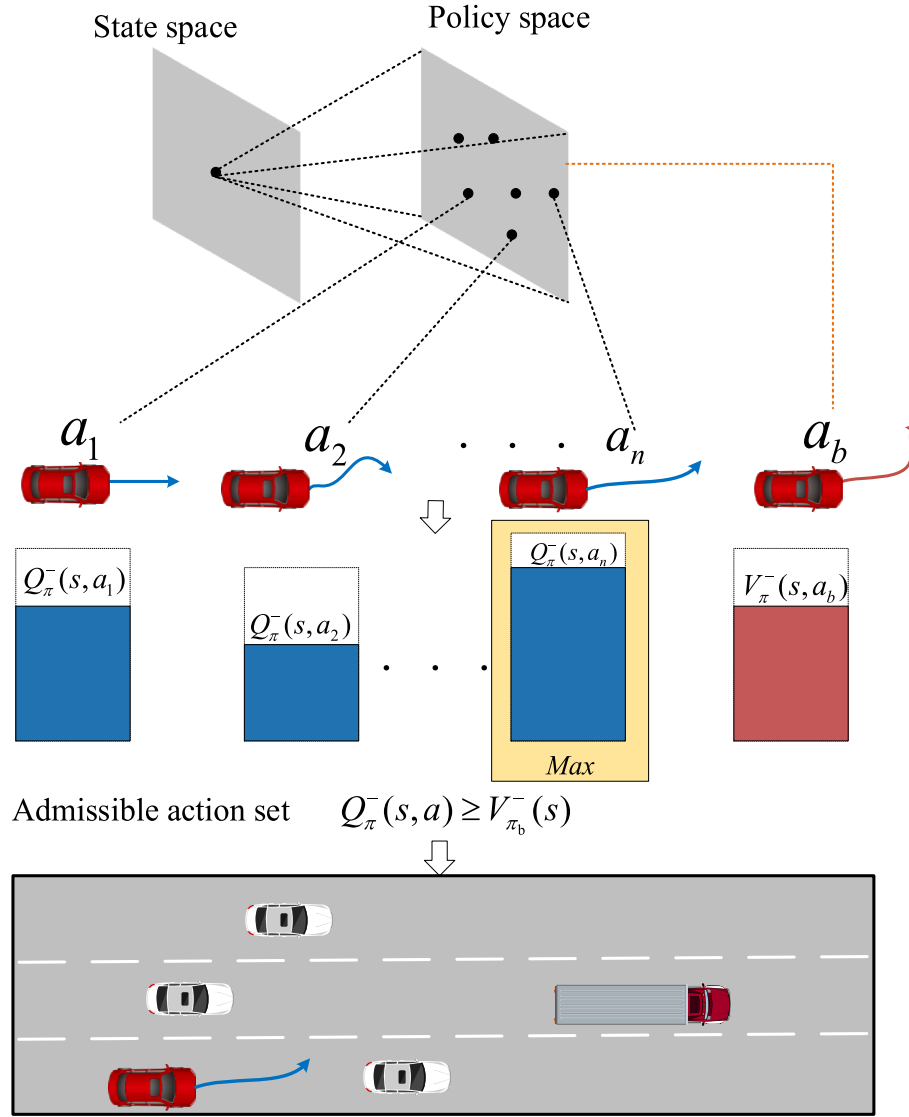


Fig. 5. Illustration of external stability.

this approach evaluates the entire set of possible control inputs to identify safe actions. To meet real-time requirements, a two-stage approach is used to compute the reachable set of the ego vehicle: 1. Offline Precomputation: The reachable set is precomputed for the initial state set $x_0 \in X_0$. 2. Online Execution: During runtime, efficiency is improved by quickly extracting the reachable subset corresponding to the current state (Kochdumper et al., 2020). For unsafe regions, the Set-Based Prediction Of Traffic Participants (SPOT) tool (Koschi and Althoff, 2017) is used to predict the possible occupation of other vehicles. The intersection of the reachable set with the unsafe regions defines the safety constraints for the control inputs. These constraints are then applied to project the control input u_a , generated by the RL policy, onto the closest safe control input u_{safe} , which is formulated as an optimization problem.

At the operational level, we adopt a zonotope polytope representation to compactly express multi-dimensional uncertainties in reachability analysis and control. The control input u is represented as:

$$u := \left\{ c_u + \sum_{i=1}^p G_{u(\cdot, i)} \alpha_i \mid \alpha_i \in [-1, 1] \right\} \quad (12)$$

which can be simplified to: $u = c_u + G_u \alpha \in \mathcal{U}$, where c_u is the center, G_u is the generator matrix, and α is a coefficient vector. For the ego

vehicle's reachable set $\mathcal{R}(t)$, we utilize the polynomial zonotope representation:

$$\mathcal{R}(t) := \left\{ c + \sum_{i=1}^h \left(\prod_{k=1}^p \alpha_k^{E_{(k,i)}} \right) G_{(\cdot, i)} + \sum_{j=1}^q \beta_j G_{I(\cdot, j)} \right\} \quad (13)$$

where α_k and $\beta_j \in [-1, 1]$. This can be simplified to: $\mathcal{R}(t) = \langle c, G, G_I, E \rangle_{PZ} \subset \mathbb{R}^n$, where c is the center, G is the generator matrix, G_I is the generator matrix for independent terms, $E \in \mathbb{N}_0^{p \times h}$ is the exponent matrix, α_k are dependent factors, and β_j are independent factors. The independent terms $\sum_{j=1}^q \beta_j G_{I(\cdot, j)}$ represent uncertainties caused by abstract errors and process noise during reachability analysis. The control input is deemed safe only if, for any possible values of the independent factors β_j , the ego vehicle's reachable set does not intersect with unsafe regions. To ensure safety throughout the entire time horizon, the safe control condition is formalized as a reachability theorem. Specifically, the ego vehicle's feasible region is defined based on the control factors α , ensuring no intersections with unsafe sets at any time $t \in [0, t_f]$. The feasible region is defined as:

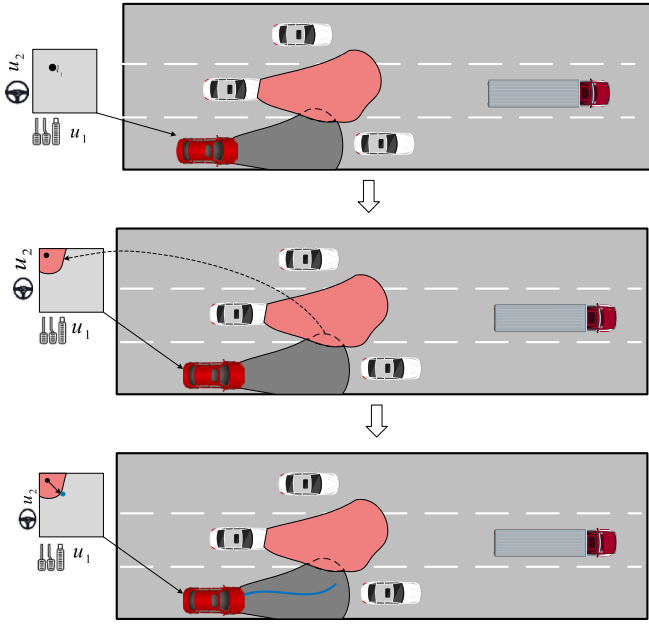


Fig. 6. Illustration of safety shield.

$$\alpha \in [-1, 1] \cap \bigcap_{r=1}^{\nu} \bigcup_{l=1}^{s_r} \underbrace{\langle a_{rl}, b_{rl}, E_{rl} \rangle_{LS}}_{\mathcal{L}S_{rl}} \quad (14)$$

where ν is the number of intersections between the reachable set and unsafe regions, s_r is the number of half-space constraints for each unsafe region, each half-space constraint $\mathcal{L}S_{rl}$ for the l constraint in the r unsafe region is defined as:

$$\mathcal{L}S_{rl} = \langle -A_{(rl,\cdot)}G, A_{(rl,\cdot)}c - \sum_{j=1}^q |A_{(rl,\cdot)}G_{I(j)}| - b_{(rl)}, E \rangle_{LS} \quad (15)$$

where $l = 1, \dots, s$. This defines a Polynomial Level Set (PLS), which describes the constraints on control inputs to ensure they do not intersect with unsafe regions at any time t .

In summary, given a factor vector $\alpha_a \in \mathbb{R}^p$, the corresponding control input u_a can be derived. To obtain a safe control input, the problem is formulated as an optimization task to project the control input onto the closest point within the safe control set. The optimization problem is formulated as follows:

$$\min_{\alpha \in [-1, 1]} \|\alpha - \alpha_a\|_1 \text{ s.t. } \alpha \in \bigcap_{r=1}^{\nu} \bigcup_{l=1}^{s_r} \langle A_{rl}, b_{rl} \rangle_p \quad (16)$$

which can be formulated as a mixed-integer linear program using Balas' Theorem (Balas, 1998):

$$\min_{\alpha \in [-1, 1]} \|\alpha - \alpha_a\|_1 \quad (17)$$

subject to the following constraints:

$$A_{rl}\hat{\alpha}_{rl} \leq \lambda_{rl}b_{rl}, -1\lambda_{rl} \leq \hat{\alpha}_{rl} \leq 1\lambda_{rl}, \lambda_{rl} \in \{0, 1\}, \alpha = \sum_{l=1}^{s_r} \hat{\alpha}_{rl}, \sum_{l=1}^{s_r} \lambda_{rl} = 1, r = 1, \dots, \nu,$$

$$l = 1, \dots, s_r, \hat{\alpha}_{rl} = \alpha_{rl}\lambda_{rl}$$

In the action projection-based safety shield, unsafe actions generated by the RL agent are corrected by the safety shield. As a result, the RL training samples will contain two types of actions: one generated by the RL agent, and the other corrected by the safety shield. If these mixed data are directly used for RL training, it may lead to instability in the

training process. Specifically, the policy used to generate training data (the behavior policy) does not fully align with the policy being learned (the target policy), thereby affecting the stability of the training (Xue et al., 2023) (Huang et al., 2024). To address this issue, we modify the loss function of the policy network by adding a safety shield guidance term, as shown below:

$$L_{\text{policy}} = \mathbb{E}_{[s_t, a_t, I(s)]} \left\{ \alpha \log \pi(s_t | a_t) - \min_{j=1, \dots, N} Q_{\theta_j}(s_t, a_t) + I(s)x_l[a_t - \mu_{\pi}(s)]^2 \right\} \quad (18)$$

The first part ($\alpha \log \pi(s_t | a_t) - \min_{j=1, \dots, N} Q_{\theta_j}(s_t, a_t)$) represents the original SAC-N loss function with the original RL action replaced by a_t that incorporates both the RL action and the one corrected by the safety shield. The second part ($I(s)x_l[a_t - \mu_{\pi}(s)]^2$) is the weighted term based on safety shield guidance. Specifically, $I(s)$ is an indicator function that marks whether a safety shield intervention occurred at state s , x_l is the weight factor for the safety shield guidance term, and $\mu_{\pi}(s)$ is the original action output by the RL policy network for state s . In this way, when the safety shield is triggered, the update direction of the policy network is guided towards a_t . For clarity, the overall process is visualized in the schematic diagram in Fig. 7, and the complete pseudocode for the proposed ARES-RL algorithm is provided in Algorithm 1 (see Table 3).

4.4. Implementation details

- 1) General Setup: In this study, all agents were trained and evaluated using the open-source CommonRoad-RL environment (Wang et al., 2021). The kinematic single-track model described in (Althoff et al., 2017) served as the vehicle dynamics model. All experiments were conducted on a computer equipped with an Intel i5-10500 processor.
- 2) RL formulation: This study formulates the decision-making and control problem as an MDP defined by observations, actions, and reward functions. The following sections provide a detailed explanation of each component.

Observations: The observations in this study are defined based on the framework described in (Althoff et al., 2017) and consist of three main components: the ego vehicle's state, road network information, and the states of surrounding traffic participants. 1. Ego vehicle state: including speed, acceleration, jerk, relative heading angle, steering angle, steering angular velocity, and yaw rate. 2. Road network information: encompassing relative offsets, lane curvature, distances to lane markings, distances to road edges, and a flag indicating whether the vehicle has exited the roadway. 3. Surrounding traffic participants: including the relative distances and velocities of up to six vehicles in the ego lane and adjacent lanes, obtained using a lane-based detection method.

Actions: In the RL framework, agents typically operate in a stepwise decision-making manner, where the policy network receives the observation from the previous time step and outputs an action for the current time step. However, decision-making and control for self-evolving learning in AVs in real-world environments must comprehensively account for dynamic changes over a period of time to effectively address challenges posed by processing delays and computation times. Moreover, to ensure safety and integrate with the online validation framework, the system needs to generate control inputs that cover the time range $[k, k + f]$. Thus, at time step k , the control inputs output by the policy network are applicable to the time range $[k, k + f]$:

$$a_{\text{RL}, k} = \begin{bmatrix} u_{\text{lon}, [k, k+f]} \\ u_{\text{lat}, [k, k+f]} \end{bmatrix} \quad (19)$$

where u_{lon} represents longitudinal control inputs, and u_{lat} represents lateral control inputs. In this study, RL-generated actions cover a temporal duration of t_f seconds, but actions are regenerated every t_c seconds, where $t_c < t_f$. This approach allows the system to adapt more flexibly to dynamic environmental changes, effectively reduce the risk of

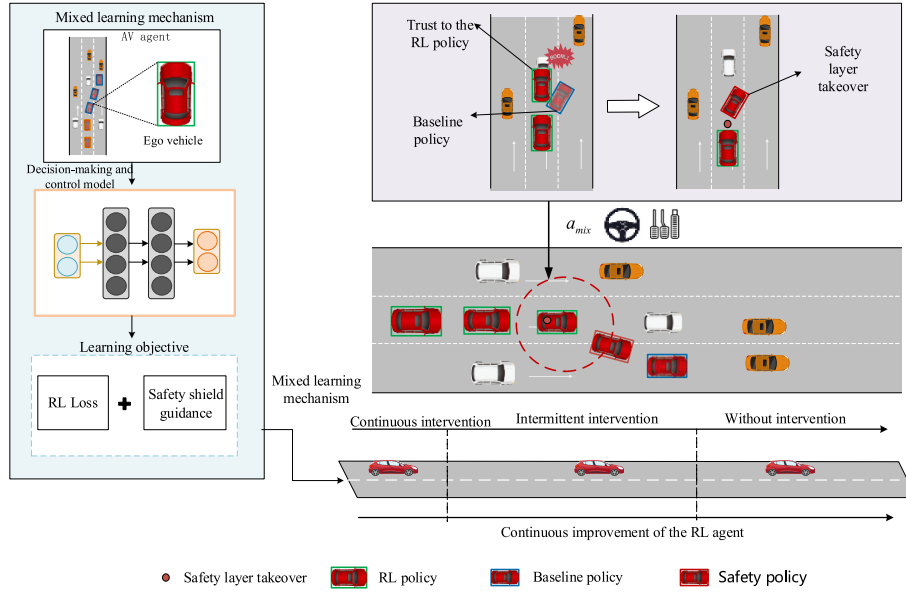


Fig. 7. The action selection process and the mixed learning process in a scenario of avoiding surrounding vehicles.

Table 3

The pseudocode of ARES-RL.

Algorithm1: An uncertainty-aware Safe-evolving Reinforcement Learning (ARES-RL)
Require: Ensemble size N , $V_{\pi_0}^-$, confidence σ and environment env
Initialize Ensemble $\{Q_{\pi}^{(j)}\}_{j=1}^N$ critic network θ and agent π , replay buffer D , Ensemble $\{Q_{\text{target}}^{(j)}\}_{j=1}^N$ target critic network θ_{int}
$s_1 \leftarrow \text{env.reset}()$
For $t = 1, 2, \dots$ do
$\overline{\mathcal{A}}(s) = \{a \in \mathcal{A} : Q_{\pi}^-(s, a) \geq V_{\pi_0}^-(s)\}$
If $\overline{\mathcal{A}}(s)$ is not empty then
$a_t \leftarrow \text{explore \& exploit}(\overline{\mathcal{A}}(s), \{Q_i\}_{i=1}^N, \pi)$
Else:
$a_t \sim \pi_{\pi}(s)$
$a_{\text{safe}} \sim \text{shield}(a_t)$
$s_{t+1}, r_t \leftarrow \text{env.step}(a_{\text{safe}})$
D.insert($s_t, a_t, r_t, s_{t+1}, I(s_t)$), where $I(s_t)$ indicates if shield was active
Update policy π using the mixed learning loss (Eq. 18),
Update policy $\{Q_{\pi}^{(j)}\}_{j=1}^N$ using gradient diversification objective (Eq. (3))
If $t \% T = 0$ and $\mathbb{P}(Q_{\pi}^-(s, a) \geq Q_{\text{target}}^-(s, a)) \geq 1 - \delta$
$\theta_{\text{int}} \leftarrow \theta$

getting stuck in deadlock situations, and improve real-time responsiveness.

Reward functions: In AD systems, the design of the reward function is critical for guiding the learning direction of the RL agent. In this study, the reward function is primarily designed based on the global objectives of the driving task, namely to maximize traffic efficiency while adhering to road speed limits and ensuring safety. A multi-objective reward function is employed, expressed as:

$$r = w_s r_{\text{safe}} + w_e r_{\text{efficient}} + w_c r_{\text{comfort}} \quad (20)$$

where w_s , w_c , and w_e are positive weight parameters for safety, efficiency, and comfort rewards, respectively. In practical settings, since safety is the top priority, w_s is set to be greater than w_e and w_c . The settings for safety, efficiency, and comfort are described as follows:

Safety reward r_{safe} : During driving, the collisions with surrounding vehicles must be avoided at all costs. If a collision occurs, a significant negative reward r_{safe1} is given. Conversely, if the ego vehicle completes the entire episode without any collisions, a large positive reward r_{safe2} is granted. The total safety reward is defined as:

$$r_{\text{safe}} = r_{\text{safe1}} + r_{\text{safe2}} \quad (21)$$

$$r_{\text{safe1}} = \begin{cases} -2, & \text{if crashed} \\ 0, & \text{otherwise} \end{cases} \quad (22)$$

$$r_{\text{safe2}} = \begin{cases} 1, & \text{if } t = T_{\text{episode}} \\ 0, & \text{otherwise} \end{cases} \quad (23)$$

Efficiency reward $r_{\text{efficient}}$: To prevent the ego vehicle from reducing collision risk by maintaining a low speed or remaining stationary, an efficiency reward related to vehicle speed is introduced. This encourages the ego vehicle to prioritize high-speed driving while ensuring safety. The efficiency reward is defined as:

$$r_{\text{efficient}} = w_v \frac{(v - v_{\min})}{v_{\max} - v_{\min}} \quad (24)$$

where $v_{\max}$ and $v_{\min}$ are the maximum and minimum speed limits of the road, and w_v is a weighting coefficient.

Comfort reward r_{comfort} : To enhance driving comfort and reduce unnecessary lane changes, a small negative reward is assigned for lane-

changing events, as defined below:

$$r_{\text{comfort}} = \begin{cases} -1, & \text{if lane change} \\ 0, & \text{otherwise} \end{cases} \quad (25)$$

5. Results and discussions

The results and discussions are organized from three perspectives: internal stability, external stability, and the integrated safety shield, to demonstrate the impact of each module on the algorithm's performance.

5.1. Internal stability

Before examining the results, we define deployment efficiency as the computation time required for each training update, including gradient updates and policy optimization. This metric is crucial in real-world learning as it directly affects whether the algorithm can operate in real time and quickly adapt to dynamic environmental changes. As shown in Table IV, a comparison between SAC-N and internal stability reveals that while maintaining similar performance, SAC-N requires an ensemble of 500 Q-networks, with each update taking approximately 1.19 s. In contrast, internal stability uses only 20 Q-networks, reducing the update time significantly to 0.048 s. Building on this analysis, Fig. 8 illustrates the training results with and without the internal stability mechanism. Specifically, when the update intervals are short (e.g., 10), although the curve exhibits an overall upward trend, frequent updates cause excessive short-term fluctuations, preventing the policy from converging towards a more stable and efficient trajectory, which in turn negatively affects the overall learning speed. When the update intervals are long (e.g., 30 or 100), the policy performance also exhibits significant instability during training, characterized by drastic fluctuations across multiple training episodes. The underlying issue may be that extended update intervals cause policy updates to diverge from the true optimization objective. The resulting accumulated deviations lead to significant performance fluctuations, making it challenging to maintain stability throughout the training process. When using the internal stability mechanism, the policy performance consistently shows a smooth upward trend with significantly reduced fluctuations. The primary reason for this improvement lies in the introduction of statistically constrained policy updates by the proposed internal stability mechanism. This mechanism utilizes an ensemble critic network to assess performance differences between the current and target policies, ensuring updates occur only under high-confidence conditions. This high-confidence condition is controlled by the parameter δ , which functions as a well-defined risk tolerance level. It specifies the maximum probability of making an erroneous update that we are willing to accept. For instance, in this paper, we set δ to 0.1, which translates to a requirement that the system must be at least 90 % confident that the new policy is an improvement. By doing so, it prevents performance degradation arising from improperly configured update intervals, thereby significantly enhancing the stability of the training process.

5.2. External stability

Fig. 9 compares the performance in terms of cumulative rewards between the policy combining external stability (red curve) and the policy using only internal stability (blue curve). It is shown that the policy integrating both internal and external stability consistently achieves higher cumulative rewards throughout the learning process and

Table 4

Deployment efficiency of SAC-N and internal stability.

	N	Deployment Efficiency
SAC-N	500	1.19s
Internal Stability	20	0.048s

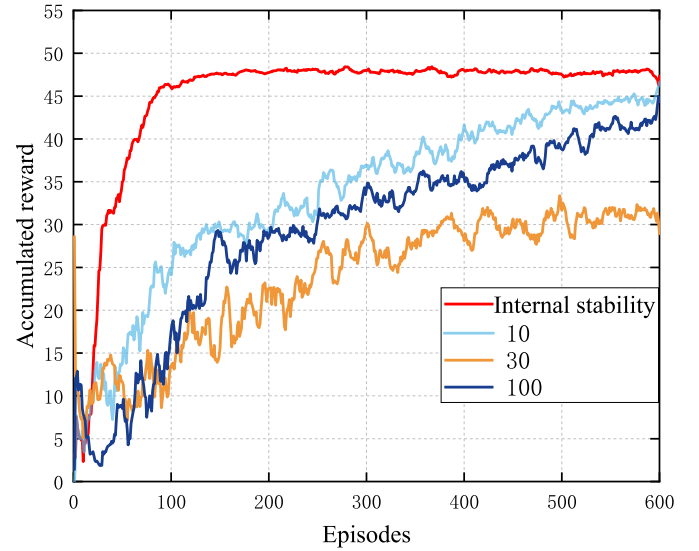


Fig. 8. Illustrates the training results with and without the internal stability mechanism.

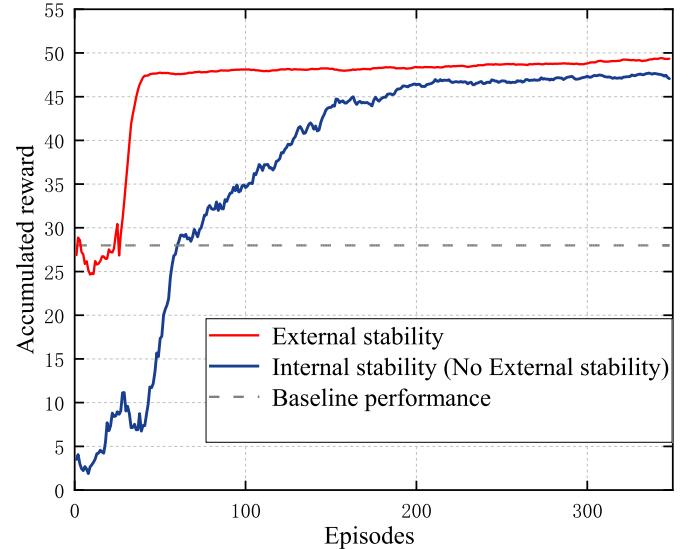


Fig. 9. Comparison of cumulative rewards between the policy combining external stability and the policy using only internal stability.

ultimately converges to better performance. The main reason the policy combined with external stability achieves faster cumulative reward improvement is its use of the benchmark policy as a reliable reference for optimizing the current policy. During the early stages of training, the use of the benchmark policy enables the system to collect high-quality data. This data provides a solid foundation for the RL algorithm to learn quickly and efficiently master critical behavior patterns, accelerating the improvement in cumulative rewards. As training progresses, the synergy between external and internal stability not only accelerates policy performance improvement in the early stages of optimization but also maintains stability during later updates, achieving a robust and continuous optimization process. In contrast, the policy using only internal stability may fall into local optima due to the lack of external references, resulting in lower cumulative rewards compared to the policy incorporating external stability. Fig. 10 illustrate the evolution of the LCB of the current policy's Q-value compared to the LCB of the benchmark policy's V-value across different training episodes. The results show that in the early stages of training, the current policy's LCB of

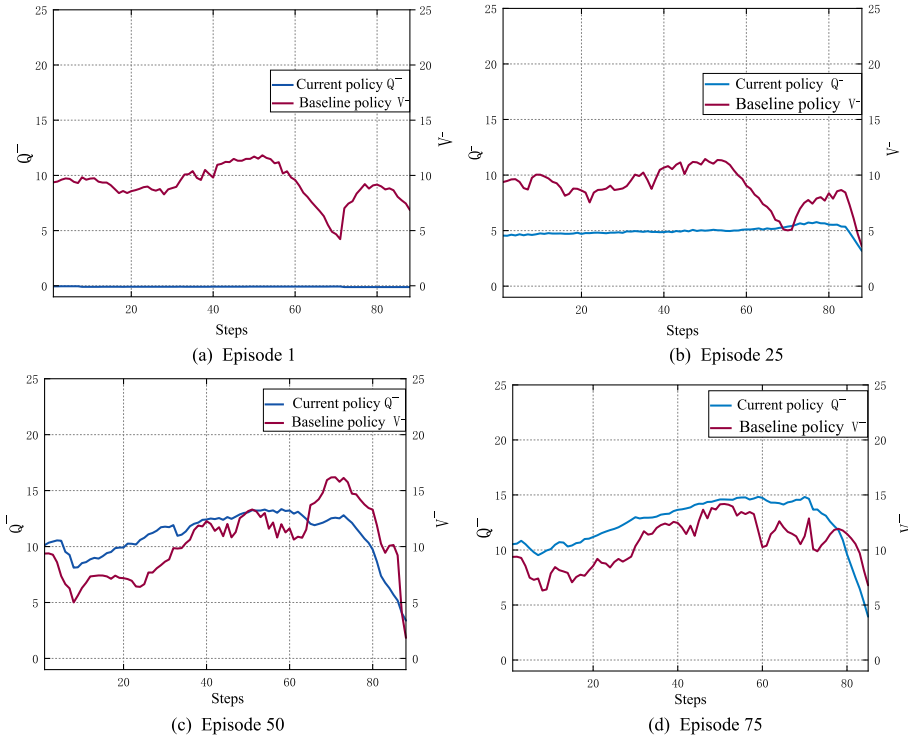


Fig. 10. Comparison of the current policy's Q-value Lower Confidence Bound (LCB) and the benchmark policy's V-value across different training episodes: (a) Episode 1, (b) Episode 25, (c) Episode 50, and (d) Episode 75.

the Q-value is lower than that of the benchmark policy's V-value, indicating that the system primarily relies on the benchmark policy for decision-making. As more data is accumulated, the current policy's LCB of the Q-value gradually surpasses the benchmark's LCB of V-value. This demonstrates a smooth transition where the system shifts from relying on the benchmark policy to optimizing and improving its own policy. This process validates the guiding role of the external stability mechanism during initial training and its contribution to enhancing policy performance.

Fig. 11 compares the two mechanisms from the perspective of safety. The cumulative unsafe events metric, which is defined as below, is used to count the total number of unsafe incidents including collisions and

off-road deviations during training:

$$\text{cumulative unsafe events}_t = \sum_{k=1}^t I(\cdot) \quad (26)$$

where $I(\cdot)$ equals 1 if a collision or off-road deviation occurs in episode k , and 0 otherwise. The results indicate that the cumulative unsafe events associated with the external stability (red curve) are significantly fewer than those of the internal stability only (blue curve). Specifically, in the early stages of training, the policy relying solely on internal stability is more prone to unsafe events, as evidenced by the rapid rise of the blue curve. In contrast, the policy combining external stability effectively reduces the risk due to the guidance of the benchmark policy. However, it should be noted that the mechanism of internal stability + external stability still shows a small number of unsafe events. For systems with extremely high safety requirements, such as AVs, even a minimal number of unsafe events is unacceptable.

5.3. Integrating safety shields

Fig. 12 visualizes the function of the safety layer during a lane-changing maneuver in the 25th training episode by comparing the state of the safety-shielded agent (yellow vehicle) with the state that would have resulted from the uncorrected RL policy (purple vehicle). This scenario is governed by our receding horizon control scheme, where actions are planned over a prediction horizon $t_f = 0.24s$ and re-evaluated at a higher frequency with a re-planning period $t_c = 0.12s$. At the initial state ($t = 0s$ (a)), both are identical. At $t = 1.2s$ (b) and $t = 2.8s$ (c), the action proposed by the original RL policy is shown to be unsafe, leading to a potential collision. In these instances, the safety layer intervenes by minimally modifying the action. This results in the yellow vehicle adopting a slightly more conservative but safe state. By $t = 4.4s$ (d), the purple vehicle is no longer visualized, indicating that the action generated by the RL policy at this time step is inherently safe and requires no correction. Consequently, the safety shield does not intervene, and the executed action is identical to the one proposed by the RL

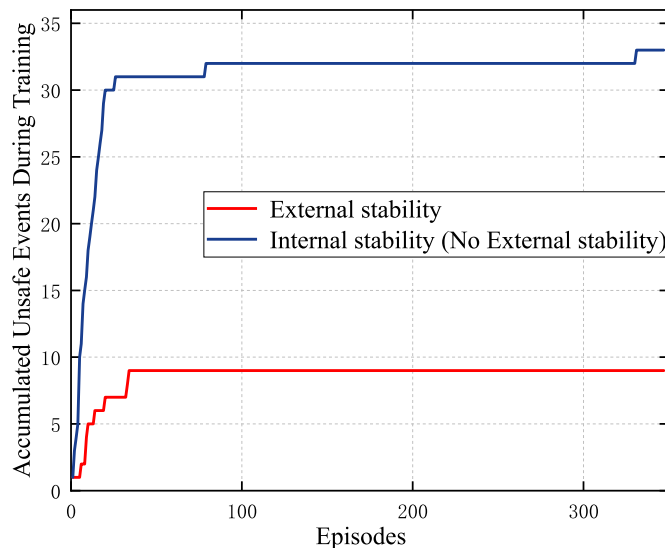


Fig. 11. Comparison of cumulative unsafe events during training between the policy combining external stability and the policy using only internal stability.

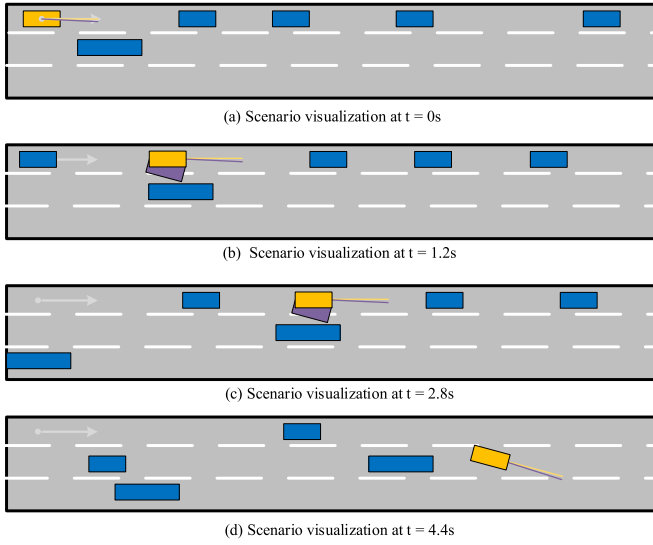


Fig. 12. Scenario visualization at different timestamps during training episode 25: (a) 0s, (b) 1.2s, (c) 2.8s, and (d) 4.4s. The purple vehicle represents the agent without the safety shield, the yellow vehicle represents the agent with the safety layer, and the blue vehicles represent other traffic participants. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

policy. This demonstrates the safety shield's ability to selectively apply minimal corrections only when necessary to guarantee safety. Figs. 13 and 14 show the performance of the policies with and without the safety shield in terms of cumulative rewards and cumulative unsafe events, respectively. In Fig. 13, the safety shield with mixed learning (red curve) overall achieves cumulative rewards higher than the benchmark policy, indicating that introducing the safety shield allows the policy to continuously exceed baseline performance. However, compared to the internal and external stability (blue curve), the safety shield with mixed learning (red curve) exhibits slower growth in cumulative rewards during the early stages of training. This is primarily due to the safety shield's constraints on high-risk actions, making the strategy more conservative in the early phase but ensuring safety. Fig. 14 shows that the cumulative unsafe events for the safety shield with mixed learning (red curve) remain at zero throughout the training process,

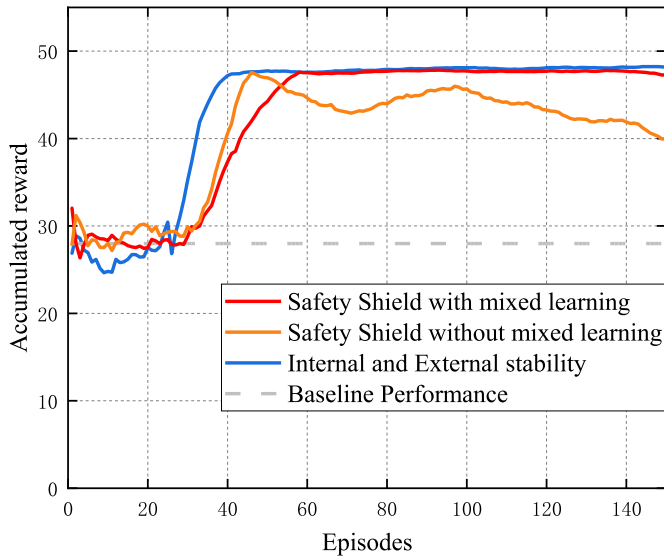


Fig. 13. Cumulative rewards comparison between policies with and without the safety shield.

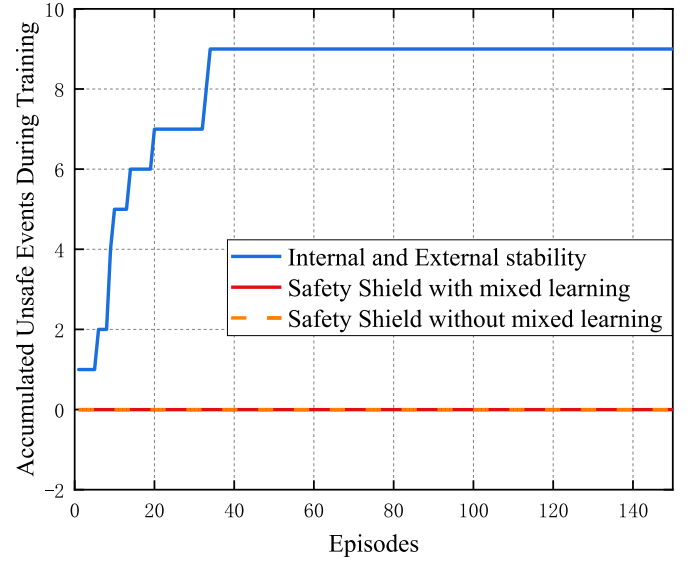


Fig. 14. Cumulative unsafe events comparison during training between policies with and without the safety shield.

demonstrating that the safety shield completely prevents unsafe behaviors during training. In contrast, the internal and external stability (blue curve) exhibits a rapid increase in cumulative unsafe events in the early stages, indicating that the policy without the safety shield is more prone to high-risk behaviors at the beginning of learning. Further analysis of the impact of the safety shield on the RL learning process reveals that, as shown in Fig. 13, without the mixed learning mechanism, the training curve declines after an initial period of growth. To investigate the cause of this phenomenon, we assess the RL policy every 10 training episodes to verify whether it has genuinely mastered the task and examine the frequency of safety shield interventions throughout the learning process. As illustrated in Fig. 15, the absence of the mixed learning mechanism results in stagnant accumulated rewards, signifying that the RL agent has not effectively learned the task. The increase in accumulated rewards in Fig. 13 is merely an illusion caused by the safety shield, which makes it appear as though the RL agent has learned. This is further supported by the increasing number of safety shield interventions shown in Fig. 16. In contrast, the proposed mixed learning

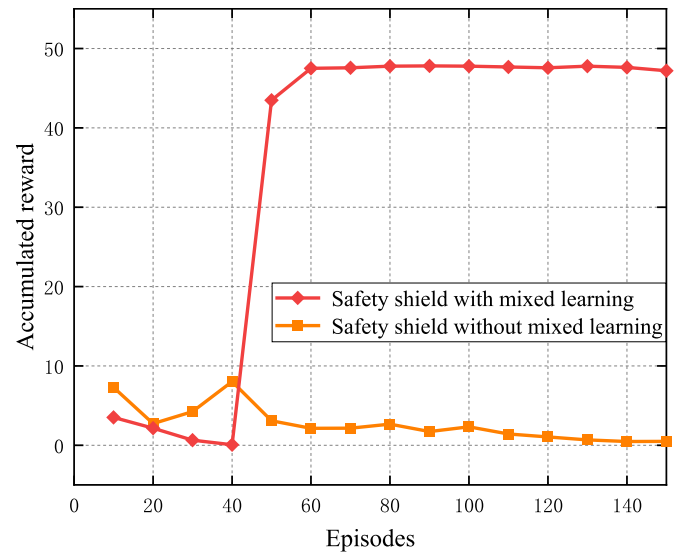


Fig. 15. Cumulative reward evaluated after every 10 training episodes with and without mixed learning procedure.

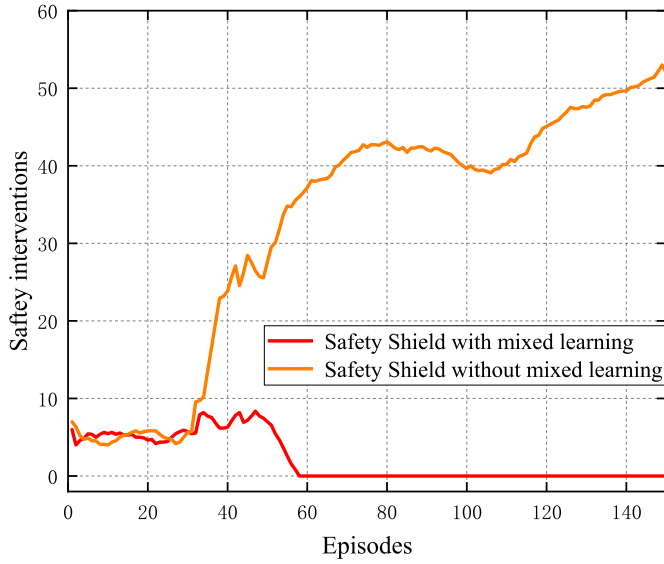


Fig. 16. Safety shield interventions with and without mixed learning procedure.

mechanism efficiently incorporates the experience of safe actions corrected by the safety shield, steering the RL agent toward learning safer and more effective behaviors, ultimately enabling it to master the task. This is achieved by setting the guidance weight x_f to 1.0, which provides a balanced objective for the agent to learn from the provided safe experience while still optimizing the original reward-based policy. To delve deeper into the reasons behind this phenomenon, Fig. 17 illustrates the trend of the LCB of the Q-value over time steps at different training episodes, comparing the performance with and without the mixed learning mechanism. The analysis reveals that in the first and

25th episodes, there is no significant difference in the Q-values between the two mechanisms. However, after the 50th episode, the Q-value for the agent without mixed learning becomes significantly higher than that of the agent with mixed learning. This phenomenon arises because the samples used for RL training contain two types of actions: those directly generated by the RL policy and those corrected by the safety shield. The use of this mixed data can lead to instability in the training process. Specifically, the policy used to generate training data (the behavior policy) does not fully align with the policy being learned (the target policy). This mismatch results in a distributional shift in the encountered state-action pairs, which in turn can lead to an erroneous overestimation of the Q-values. This overestimation causes the policy to become overconfident in its current actions (even if they are poor), which in turn explains the increasing frequency of safety shield interventions observed in Fig. 16.

Additionally, the collaborative learning process between the safety shield and the RL agent is analysed in Fig. 18, which examines both the number of interventions per episode (blue curve) and the average size of the admissible action space per episode (red curve) during training. In the early stages of training, since the AD policy lacks experience, the admissible action space is zero, and the policy relies entirely on the benchmark policy for guidance. Meanwhile, the number of safety interventions remains relatively low, indicating that the policy primarily follows the benchmark policy's guidance during this phase. As training progresses, the policy gradually learns and begins to improve its capacity for independent decision-making. The admissible action space (red curve) expands rapidly, and the number of safety interventions (blue curve) also increases, reaching a peak. This increase in interventions can be attributed to the balance between exploration and safety. Subsequently, as the policy further improves and internalizes the benchmark policy's safe behaviors, the number of safety interventions begins to decrease, and the admissible action space stabilizes. This indicates that the current policy has significantly improved its ability to make AD within the safe range, gradually reducing its reliance on the

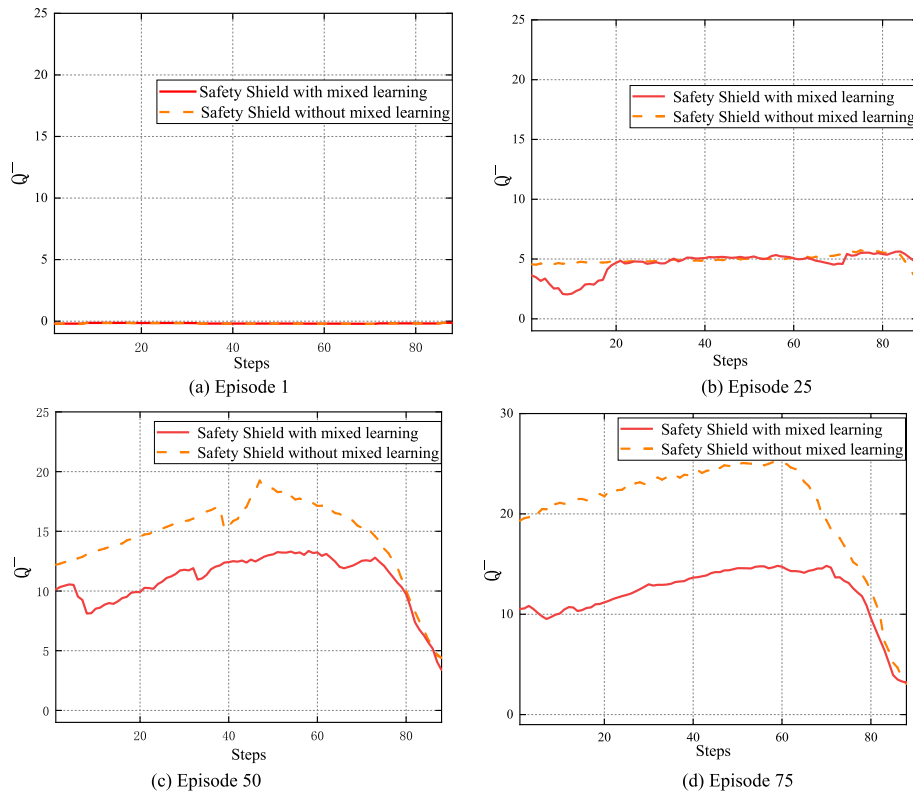


Fig. 17. Evolution of the Q-value Lower Confidence Bound (LCB) over time steps with and without the mixed learning mechanism, shown at different training episodes: (a) Episode 1, (b) Episode 25, (c) Episode 50, and (d) Episode 75.

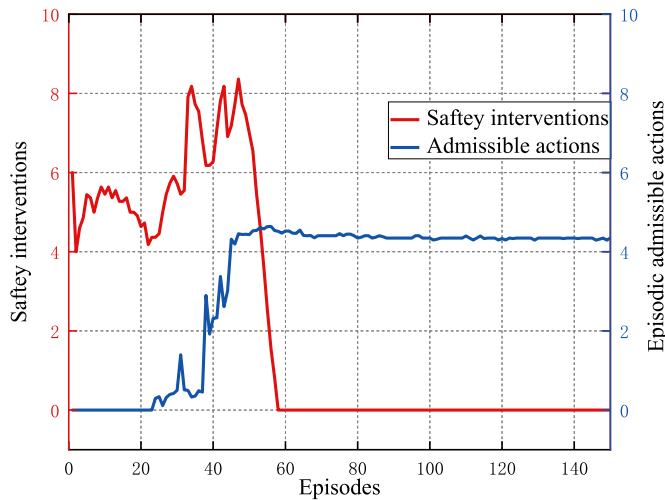


Fig. 18. Analysis of safety shield interventions and the size of the admissible action space during training.

safety shield. In the later stages of training, the number of safety interventions approaches zero, indicating that the policy can reliably make decisions entirely within the safe range.

5.4. Comparison with baseline algorithms

To assess the effectiveness of our algorithm, we conducted a comparison with the PPO (Schulman et al., 2017) and SAC (Haarnoja et al., 2018) algorithms. As illustrated in Fig. 19, PPO and SAC experience instability during training, particularly in the early stages. In contrast, the proposed algorithm ensures steady improvement in cumulative rewards and achieves higher training efficiency by effectively leveraging both internal and external stability mechanisms. Fig. 20 further compare the performance from a safety perspective. The red curve indicates that the integrated safety algorithm maintains zero cumulative unsafe events throughout the training process, demonstrating that the safety mechanisms effectively prevent unsafe behaviors such as collisions or off-road deviations. Conversely, PPO and SAC algorithms accumulate a significant number of unsafe events in the early stages of training, reflecting the high-risk nature of PPO and SAC algorithms during the initial exploration phase.

6. Conclusion

This paper proposes an uncertainty-aware safe-evolving RL algorithm and validates its effectiveness in dynamic and uncertain highway lane-changing scenarios. The algorithm incorporates an internal stability mechanism aimed at improving the stability of policy updates themselves, facilitating learning in a direction that leads to better performance over time. Meanwhile, an external stability mechanism is introduced, using a benchmark policy as a reference to guide the optimization direction of the current policy, ensuring the algorithm consistently outperforms the baseline. Furthermore, an action projection mechanism and a mixed learning procedure are integrated to make minimal adjustments to the learned policy to ensure safety while enabling stable learning from both the safe and original actions. Simulation results demonstrate that the proposed algorithm not only achieves cumulative rewards comparable to baseline online learning algorithms such as SAC and PPO but also exhibits significant advantages in learning stability, safety, and training efficiency. Although this study focuses on addressing epistemic uncertainty, future research should explore the integration of aleatoric uncertainty (e.g., inherent randomness in the environment) to understand its impact on the learning process. This would further enhance the adaptability and robustness of the algorithm

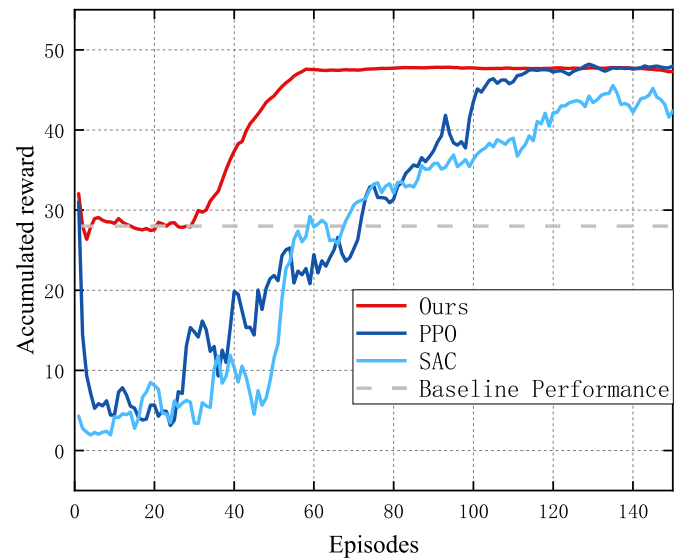


Fig. 19. Comparison of cumulative rewards between the proposed algorithm, PPO, and SAC.

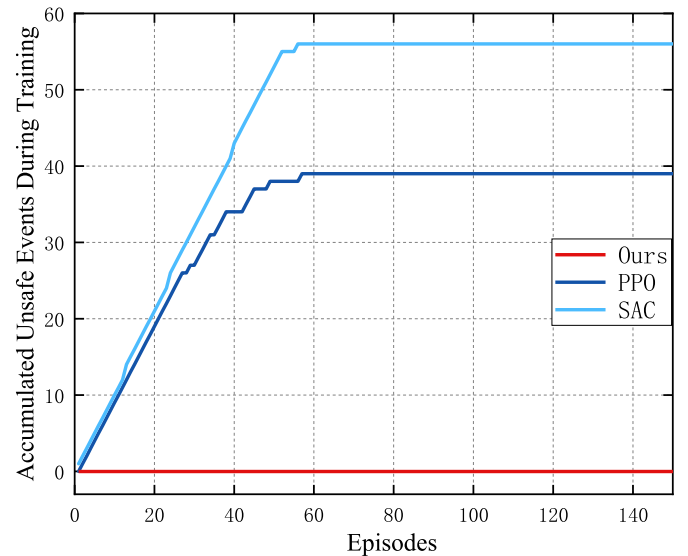


Fig. 20. Comparison of cumulative unsafe events during training between the proposed algorithm, PPO, and SAC.

in more complex and diverse scenarios.

CRediT authorship contribution statement

Ping Lu: Writing – original draft, Software, Methodology, Conceptualization. **Sunan Zhang:** Writing – review & editing, Resources, Methodology, Conceptualization. **Feihong Tan:** Writing – review & editing, Resources, Data curation. **Fulin Zhang:** Writing – review & editing, Resources. **Yuxiang Feng:** Validation, Software, Investigation, Formal analysis. **Bo Hu:** Validation, Project administration, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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